

Lab 5 report

1. How do we characterize (discretize) states of the world in this learning task?

The states of the world in this learning task are characterized by four continuous variables:

- Cart position (x): Continuous range between -2.4 and 2.4
- Cart velocity ($\dot{x}$): Continuous range between -3.0 and 3.0
- Pole angle (θ): Continuous range between -12° and 12° (converted to radians)
- Pole angular velocity ($\dot{\theta}$): Continuous range between -2.0 and 2.0.

Each range is divided into 5 intervals/bins using `np.linspace`.

`np.digitize` then maps each continuous variable to a bin index to discretize the continuous state space into a finite set of discrete states. The final discrete state is represented as a set of bin indices: (x_{bin} , $\dot{x}_{\text{bin}}$, θ_{bin} , $\dot{\theta}_{\text{bin}}$).

2. How is the reward function designed?

The reward function is designed to encourage survival and penalize failure (cart out of bounds). If the cart stays within bounds ($\text{abs}(x) \leq 2.4$) and the pole upright ($\text{abs}(\theta) \leq 12^\circ$), the agent receives a reward of 1. However, if the cart goes out of bounds ($\text{abs}(x) > 2.4$) or the pole angle limit is exceeded ($\text{abs}(\theta) > 12^\circ$), the agent receives a reward of -1 and the current episode in training ends. The reward function therefore encourages the agent to keep the pole in balance and avoid falling out of bounds for as many steps as possible.

3. What is the function to be learned?

The function to be learned is the optimal Q-value function $Q^*(s, a)$, which is the expected cumulative reward when taking the action a in state s . This function represents the optimal value of performing a specific action a in a state s when considering the future rewards. The Q-values allow the agent to choose the best action in a selected state, which maximizes the cumulative reward (optimal policy).

4. What learning algorithm is used?

The learning algorithm used is Q-learning, a reinforcement learning algorithm. It updates the Q-values using Bellman equation:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \cdot [r + \gamma \cdot \max_{a'} Q(s', a') - Q(s, a)]$$

Where s' is the next state, a' is the next action, α is the learning rate, r is the reward and γ is the discount factor.

Learning rate α is dynamically updated depending on how many times each state-action pair has been visited. Frequently visited state-action pairs will therefore have an increasingly smaller α and stabilized Q-value.

Discount factor γ is a constant (0.99) to make cumulative reward have a finite value.

5. How do you choose exploratory actions?

Exploratory actions are made using the epsilon-greedy strategy. With probability ϵ , a random action is chosen, encouraging the agent to explore new state-action pairs which might not give maximum reward. With probability $1 - \epsilon$, the action with the highest Q-value for the current state is chosen, encouraging exploitation. As epsilon ϵ decays over episodes, it favors exploitation over exploration by choosing the action with highest Q-value, allowing for maximal cumulative reward.

6. How many intervals can your system survive by using the learning strategy?

The system can at least survive over a minute (3000 steps), so we can assume that the system can survive indefinitely. These are two separate attempts which survived 3000 steps:

